SUSHMA YADAV

Tech Lead | Full-Stack | Python & GenAI Engineer

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PROFESSIONAL SUMMARY

Generative AI and Python Engineer and Tech Lead with 10 years of software engineering experience, including 2 years of **Python (FastAPI)** and 1 year of hands-on **Generative AI** work: **RAG** applications, **vector databases and embeddings** (pgvector), **LLM APIs** (Claude, Groq), LLM training for **text-to-SQL** generation, **MCP** connectors, prompt engineering, and metric-based **LLM evaluation** (ROUGE, BERTScore, LLM-as-judge). Brings production depth from building, migrating and scaling platforms in **Drupal, PHP, Laravel, Node.js and React**; leads a team of 5 engineers, directed two legacy-to-modern CMS migrations (100+ modules combined) and architected multi-tenant SaaS platforms. Hands-on with **AWS, Docker, CI/CD, PostgreSQL, Redis** and **Supabase**.

KEY HIGHLIGHTS

- **Generative AI: RAG** pipelines, **pgvector** semantic search and an in-house **FastAPI** cross-sell vector-search platform.
- **LLM applications: text-to-SQL** LLM training, **MCP connector**, **agentic web testing** and LLM **scam detection** (Groq).
- **LLM evaluation:** prompt variants scored with **ROUGE**, **BERTScore** and **LLM-as-judge**; token-usage optimization.
- **10 years** in software engineering; **Tech Lead of 5 engineers**; **2 Drupal migrations** (100+ modules, **zero data loss**).
- Architected a **multi-tenant Laravel SaaS** (Taxnomist); built a **React helpdesk** that replaced a licensed **Zendesk** subscription.
- **Stack:** Python, FastAPI, PHP, Node.js, React, AWS, Docker/Docksal, CI/CD, PostgreSQL, Redis, pgvector.

TECHNICAL SKILLS

Generative AI / LLM	RAG (Retrieval-Augmented Generation), Embeddings, Vector Databases (pgvector), Semantic Search, LLM Training, Text-to-SQL, MCP (Model Context Protocol), Agentic AI Workflows (Zapier), Text Classification, Prompt Engineering (zero-shot, few-shot), LLM APIs (Claude, Groq), Hugging Face, LLM Evaluation (ROUGE, BERTScore, LLM-as-judge), Token Optimization
Languages	Python, PHP, JavaScript, Node.js, SQL, HTML5, CSS3
Frameworks & Libraries	FastAPI, Pydantic, Laravel, React, Drupal 7 to 11, WordPress, jQuery, REST APIs
E-Commerce / CMS	Shopify, Drupal, WordPress - theme customization, app/plugin/module customization and API integrations
Databases	PostgreSQL, MySQL, Redis, Supabase, pgvector
Cloud & DevOps	AWS (EC2, S3, RDS), Docker, Docksal, Containerized Development Environments, CI/CD Pipelines, Git, Deployment Automation
Code Quality	SonarQube, Quality Gates, Code Review, Static Code Analysis
Architecture	Multi-Tenant SaaS Design, REST API Design, Legacy-to-Modern Migration, Technical Roadmapping
AI-Assisted Development	Claude, GitHub Copilot, Hugging Face, Lovable AI
Leadership	Team Leadership and Mentoring (team of 5), Client-Facing Delivery, Agile Delivery

PROFESSIONAL EXPERIENCE

Tech Lead / Software Engineer — Busywizzy Technology Pvt. Ltd. | Jun 2016 – Present

Leadership, Architecture & Delivery

- Lead and mentor a **team of 5 engineers** delivering **Drupal**, WordPress, **Laravel**, **Shopify**, **Node.js** and **React** projects for enterprise and product-based clients.
- Directed two legacy-to-modern CMS migrations (**Drupal 7 to Drupal 11** and **Drupal 7 to Drupal 10**), each covering **50+** custom and contrib modules, with **zero data loss** and no extended downtime.
- Architected a **multi-tenant Laravel** SaaS platform for tax-notice workflow management with tenant-isolated data and secure access controls.
- Led development of a custom **React**-based support and ticketing platform (**Lovable AI**-assisted) that replaced a licensed **Zendesk** subscription and cut recurring software costs.

Full-Stack & E-Commerce

- Built and customized **Shopify** storefronts, including theme development with **Liquid** and integrations with third-party apps and **Shopify** APIs.
- Built **Node.js** services and REST APIs for integrations between front-end applications, third-party tools and backend platforms.

Python, AI & Data

- Developed **Python** REST APIs with **FastAPI** (**Pydantic** validation, auto-generated **OpenAPI** docs) for in-house AI and vector-search projects.
- Designed **PostgreSQL** and **MySQL** schemas and optimized queries; used **Redis** and **Supabase** in new projects.
- Introduced AI-assisted development (Claude, GitHub Copilot, Hugging Face) across the team to speed up delivery and improve code quality.
- Built in-house **Python** Generative AI projects, including an LLM-powered customer-complaint response generator using **few-shot** prompting, comparing prompt variants with **ROUGE**, **BERTScore** and **LLM-as-judge**.
- Developed embedding-based semantic search and **RAG** workflows using a vector database (**PostgreSQL** with **pgvector**) and LLM APIs to answer questions over project documentation and support content.

Cloud, DevOps & Code Quality

- Deployed and operated applications on **AWS** (**EC2**, **S3**, **RDS**), covering environment setup, file storage and managed database hosting.
- Set up **Docker** and **Docksal**-based containerized development environments for **Drupal**, WordPress and **Laravel** projects, giving the team consistent, reproducible local setups that match staging and production.
- Set up and maintained **CI/CD** pipelines to automate build, test and deployment, reducing manual release effort and deployment errors.
- Integrated **SonarQube** quality gates to track bugs, code smells and security hotspots, improving maintainability of PHP and JavaScript codebases.

SEO/SMO Executive — Webart Solution Pvt. Ltd. | Mar 2013 – Jul 2015

- Built foundational web and digital-marketing skills before transitioning into software engineering.

FEATURED PROJECT: E-COMMERCE CROSS-SELL VECTOR SEARCH (IN-HOUSE)

Product Cross-Sell Recommendation Platform (Python, FastAPI, PostgreSQL, pgvector, Embeddings, Docker)

- Built an in-house platform that suggests complementary products for e-commerce cross-sell using semantic vector search instead of keyword or manual rule matching.
- Generated text embeddings from product titles, descriptions and attributes, and stored them in **PostgreSQL** using the **pgvector** extension.
- Developed **FastAPI** REST endpoints for product ingestion, embedding generation and top-N similar and complementary product retrieval using cosine similarity.
- Designed the **PostgreSQL** schema and vector indexing for approximate nearest-neighbour search across the product catalog.
- Validated request and response models with **Pydantic** and exposed interactive **OpenAPI** (Swagger) documentation for API consumers.
- Containerized the service with **Docker** and tested recommendation relevance with sample product queries.

AI ENGINEERING PROJECTS

- **RAG Application (In-house):** Retrieval-Augmented Generation pipeline combining vector search with LLM APIs (Groq) to produce grounded answers from internal content.
- **SQL Query Generator (In-house):** worked on training an LLM for an internal system that generates SQL queries from natural-language questions.
- **Vector Database and Embeddings:** generated text embeddings and stored them in pgvector for semantic similarity search over documents.
- **LLM Dataset Analysis:** analyzed datasets for token optimization to improve LLM response quality and reduce usage.
- **Customer-Complaint Response Generator:** Python few-shot prompting application with prompt variants evaluated using ROUGE, BERTScore and LLM-as-judge.
- **Scam Detection System:** AI-powered Python application that classifies text messages as Scam, Not Scam or Uncertain using the Groq LLM API.
- **MCP Connector:** built a Model Context Protocol (MCP) connector for an in-house internal project management system, exposing project and task data to AI assistants.
- **Agentic Web Testing Application:** built with Zapier for a testing platform; an agentic workflow interprets the user's query and performs the corresponding tests on the target website.
- **Software Q&A Chatbot:** built a chatbot on an OpenAI model that answers only software-related questions; manages chat history and uses query rewriting to resolve follow-up questions, with token optimization and prompt evaluation to scale.
- **Prompt Engineering:** applied hands-on to accelerate code development and improve engineering efficiency.

SELECTED PROJECTS

RiseSmart — Recruitment SaaS Platform (Drupal, Docksal) | cms.risesmart.com

- Led migration of a live product platform from **Drupal 7 to Drupal 11**, modernizing **50+** custom and contrib modules; ran development and testing in a **Docksal** (**Docker**) environment.

EthosCE — Enterprise Learning Management Platform (Drupal, Docksal) | gocadmium.com/ethosce

- Migrated a production LMS from **Drupal 7 to Drupal 10**, modernizing **50+** modules while preserving course data and continuing-education compliance records; used a **Docksal** (**Docker**) environment for development and testing.

SBS Texas — Support and Ticketing Platform (React, Lovable AI) | silveradobusinesssystems.com

- Built a **Zendesk**-equivalent helpdesk management system (ascent.sbstexas.com) from scratch, eliminating recurring third-party licensing costs.

Taxnomist — Multi-Tenant Tax Workflow Platform (Laravel, Docker) | taxnomist.wizenglobal.com

- Designed **multi-tenant** architecture (currently serving 2 client tenants) for tax notice and acknowledgement workflows with tenant-isolated client data; containerized the project setup with **Docker**.

EDUCATION

M.Tech, Computer Science — Maharishi Dayanand University, 2015 | 77%
B.Tech, Computer Science — Maharishi Dayanand University, 2013 | 79%